

CLASSIFICATION

UNIT 3

DISCLAIMER

The contents are adapted from various internet sources such as:

1. Andrew NG Stanford course
2. SimpliLearn.com
3. Coursera.org
4. Analytics Vidhya.com

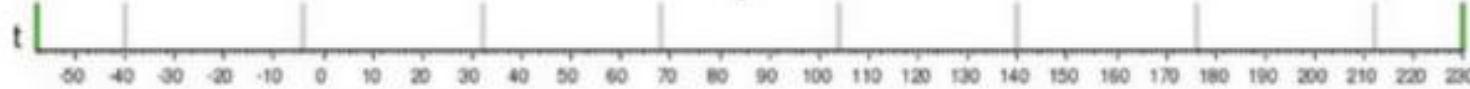
RECAP

Regression



What will be the temperature tomorrow?

84°



Fahrenheit

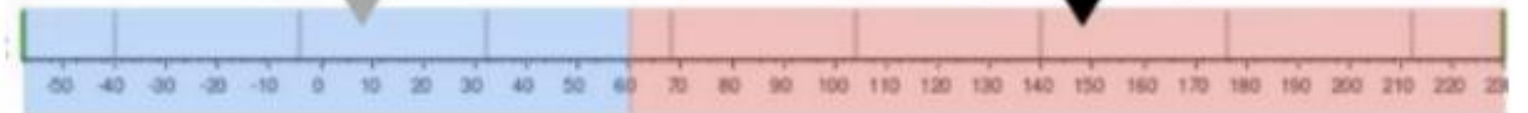
Classification



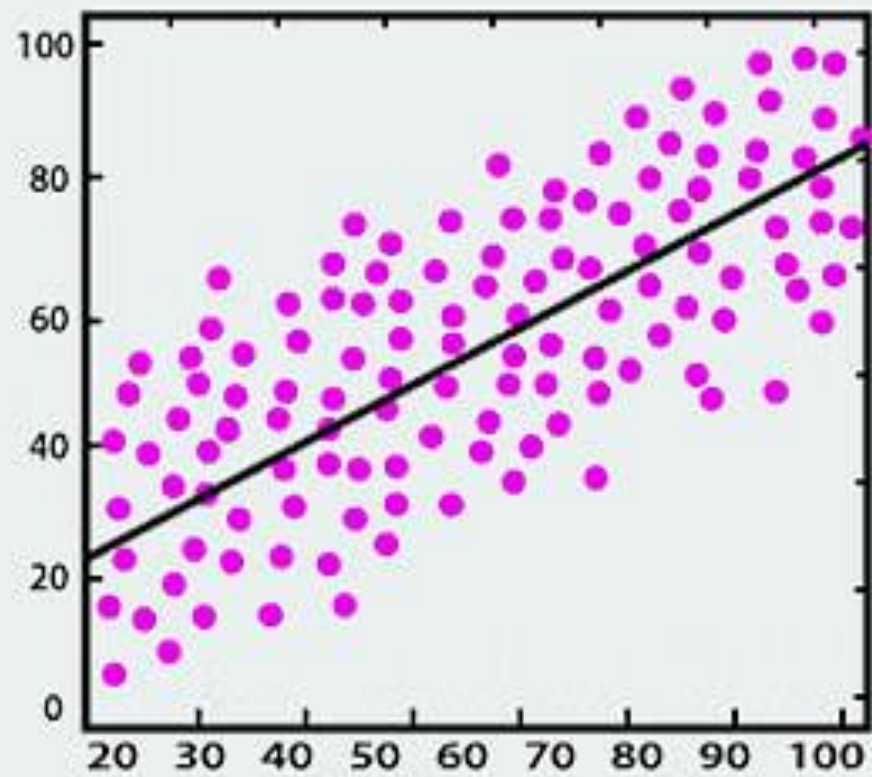
Will it be hot or cold tomorrow?

COLD

HOT

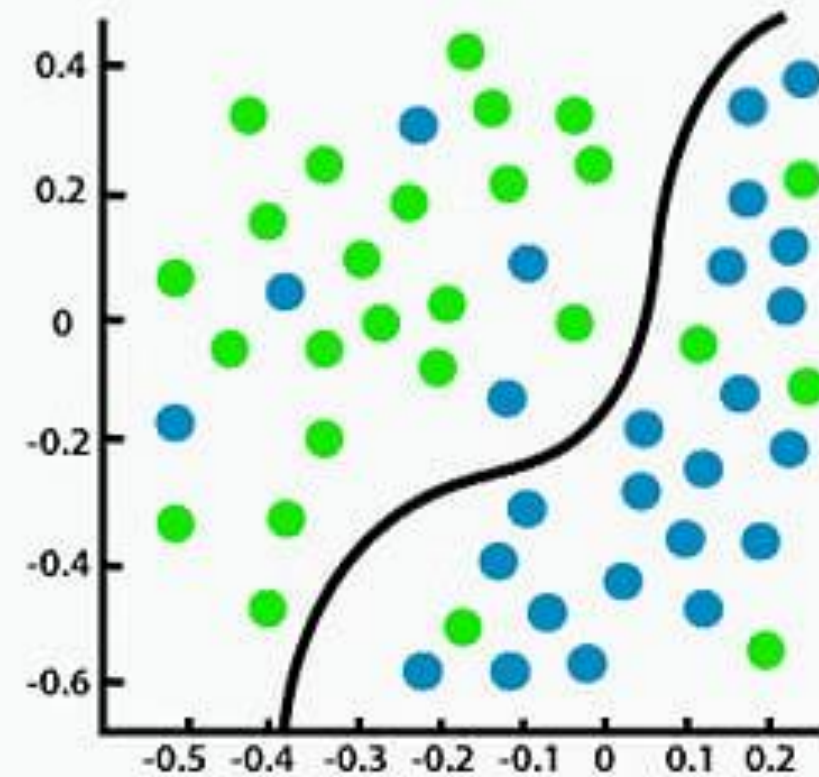


Fahrenheit



Regression

versus



Classification

Regression

- Supervised Learning
- Output is a continuous quantity
- Main aim is to forecast or predict
- Eg: Predict stock market price
- Algorithm: Linear Regression

Classification

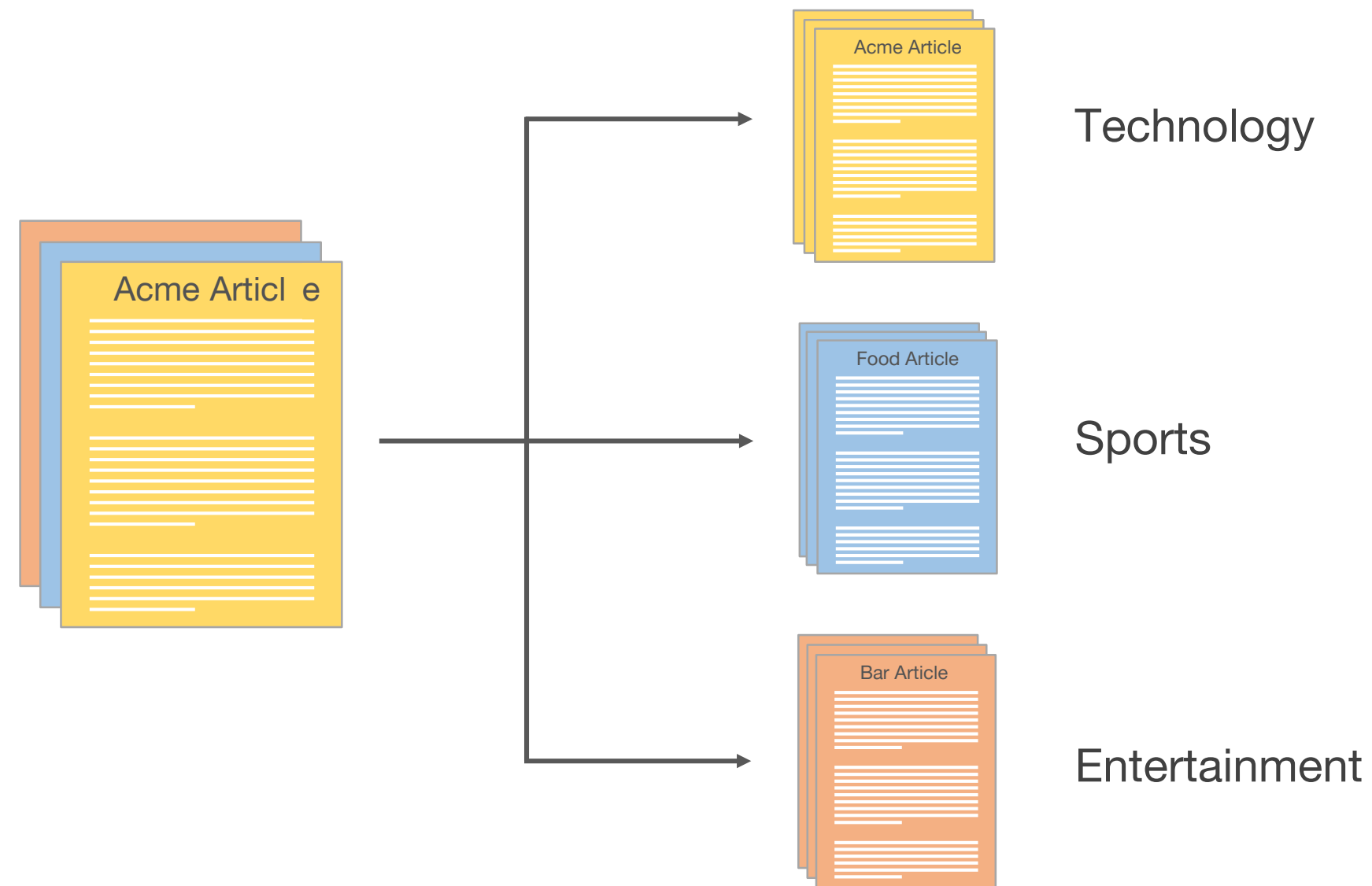
- Supervised Learning
- Output is a categorical quantity
- Main aim is to compute the category of the data
- Eg: Classify emails as spam or non-spam
- Algorithm: Logistic Regression

Clustering

- Unsupervised Learning
- Assigns data points into clusters
- Main aim is to group similar items clusters
- Eg: Find all transactions which are fraudulent in nature
- Algorithm: K-means

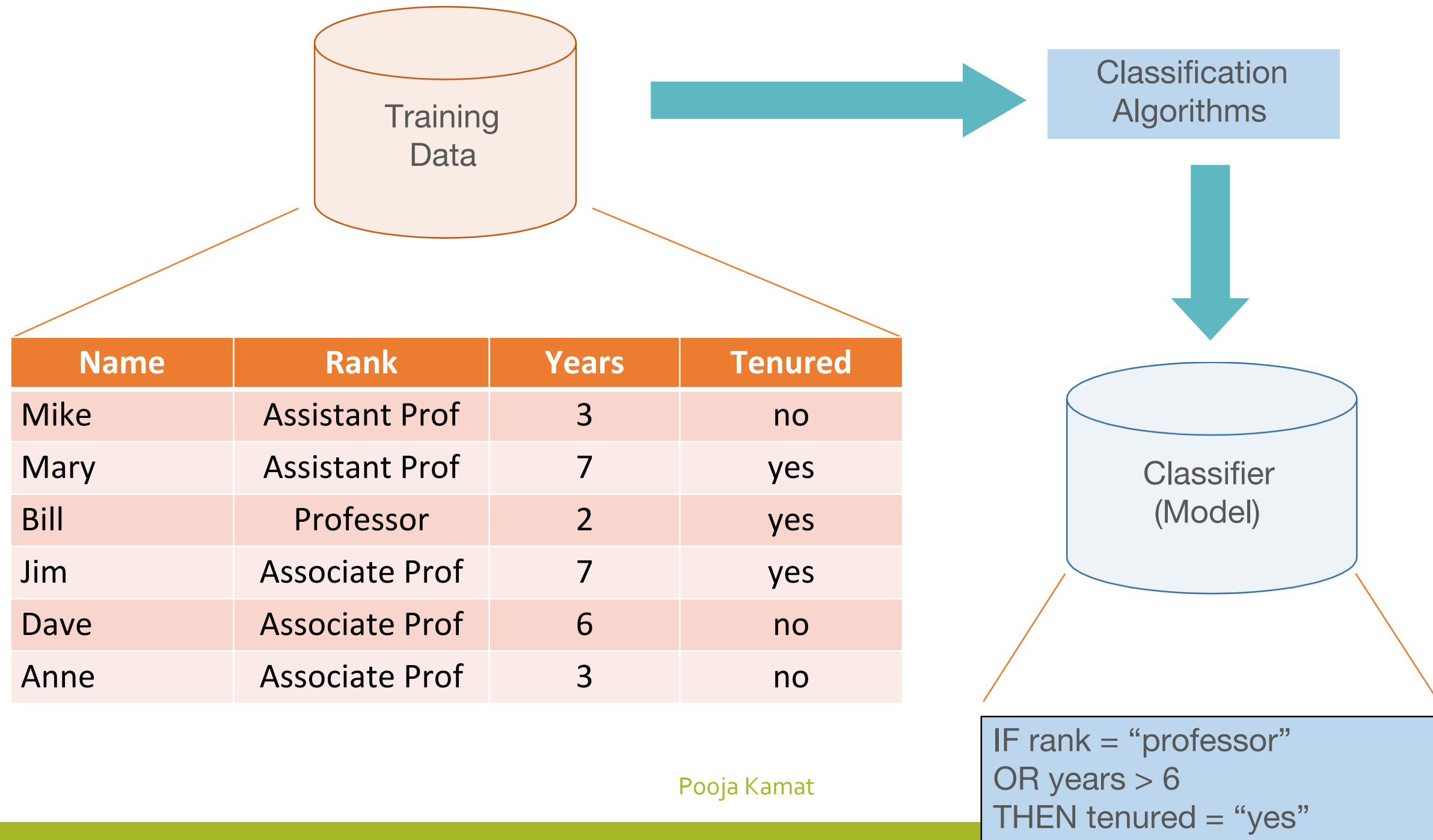
What Is Classification?

A machine learning task that identifies the class to which an instance belongs



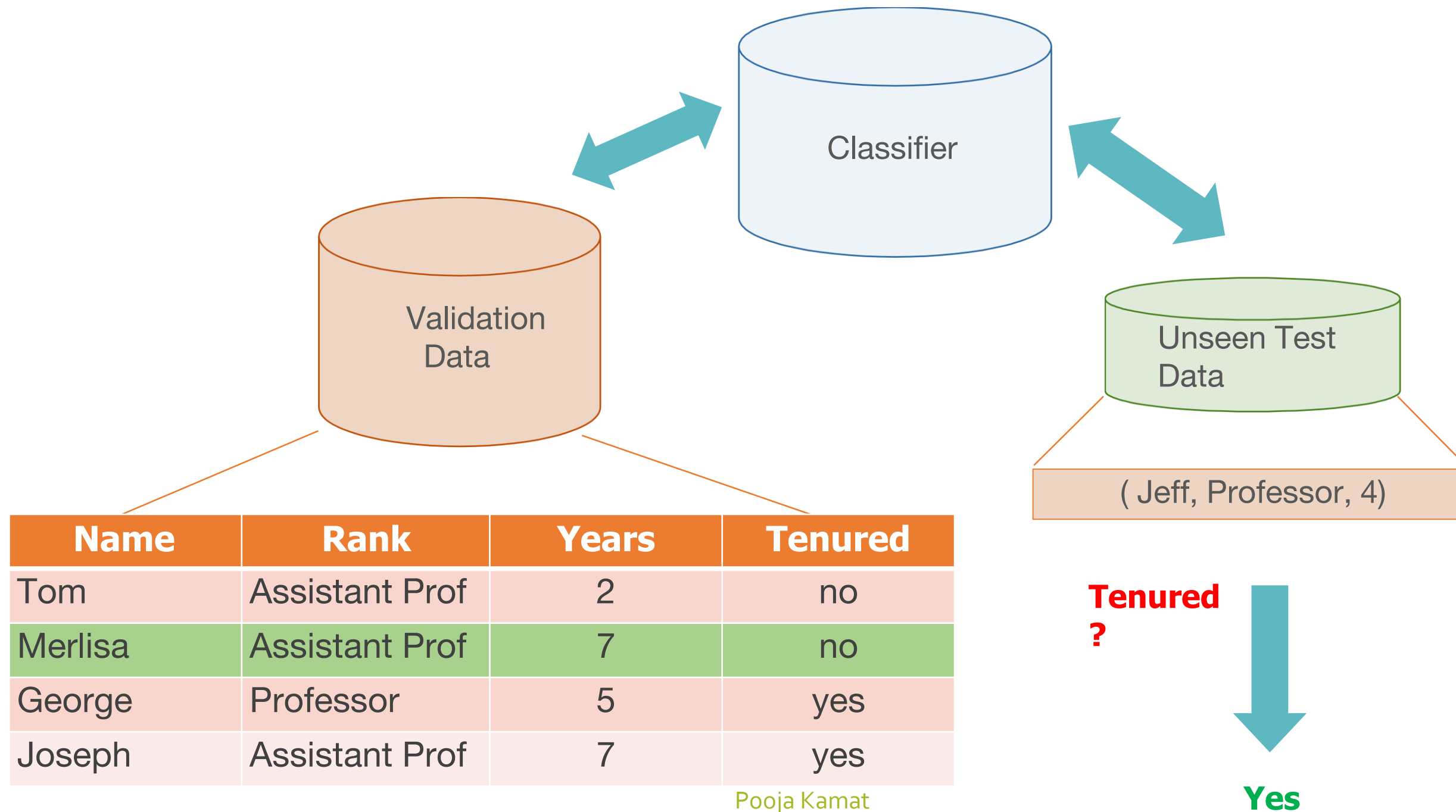
Classification: Example

Training a classifier model with respect to the available data



Classification: Example

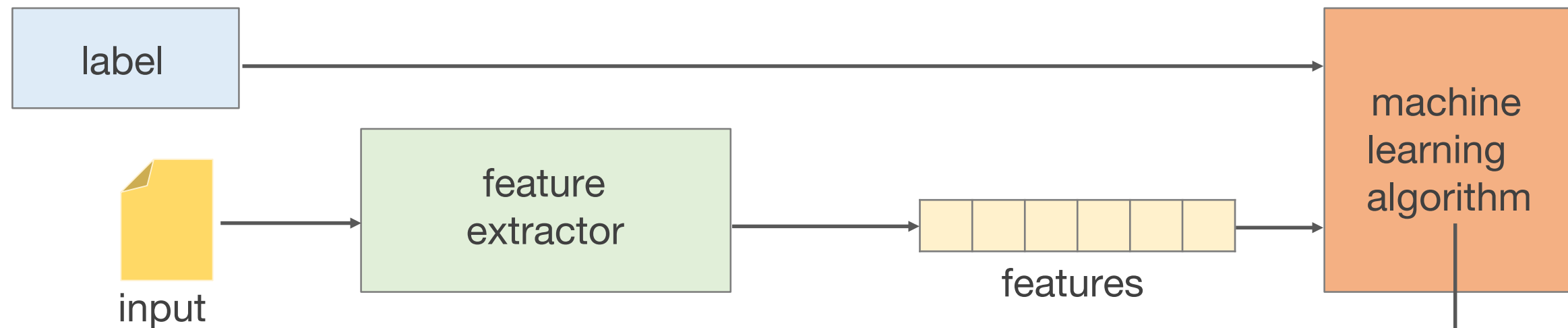
The model will classify if the professors are tenured or not



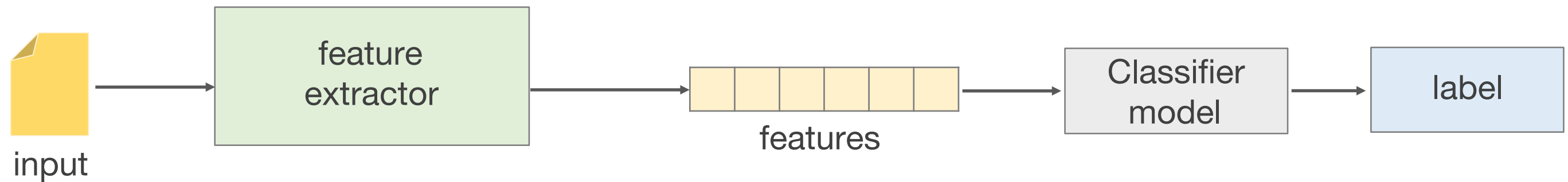
Pooja Kamat

Classification: Work Flow

(a) Training



(a) Prediction

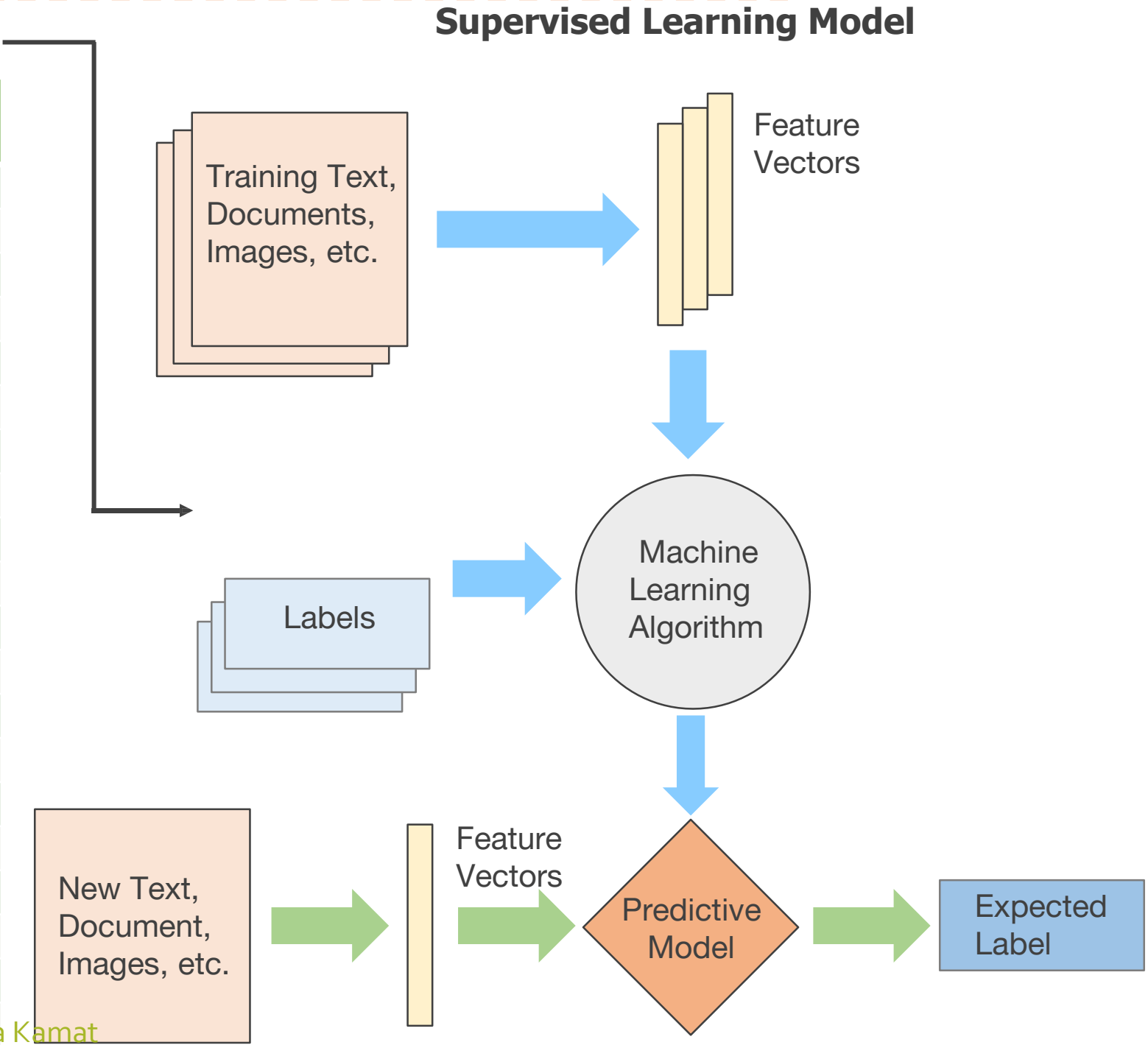


Original Source: Simplilearn.com

Classification: A Supervised Learning Algorithm

Classification is a supervised learning algorithm as the training data contains labels

Record ID	Age	Spectacle Prescription	Astigmatic	Tearproduction Rate	Class Label Lenses
1	Young	Myope	No	Reduced	Noncontact
2	Young	Myope	No	Normal	Soft contact
3	Young	Myope	Yes	Reduced	Noncontact
4	Young	Myope	Yes	Normal	Hard contact
5	Young	Hypermetrope	No	Reduced	Noncontact
6	Young	Hypermetrope	No	Normal	Soft contact
7	Young	Hypermetrope	Yes	Reduced	Noncontact
8	Young	Hypermetrope	Yes	Normal	Hard contact
9	Pre-presbyopic	Myope	No	Reduced	Noncontact
10	Pre-presbyopic	Myope	No	Normal	Soft contact
11	Pre-presbyopic	Myope	Yes	Reduced	Noncontact
12	Pre-presbyopic	Myope	Yes	Normal	Hard contact
13	Pre-presbyopic	Hypermetrope	No	Reduced	Noncontact
14	Pre-presbyopic	Hypermetrope	No	Normal	Soft contact
15	Pre-presbyopic	Hypermetrope	Yes	Reduced	Noncontact
16	Pre-presbyopic	Hypermetrope	Yes	Normal	Noncontact
17	Presbyopic	Myope	No	Reduced	Noncontact
18	Presbyopic	Myope	No	Normal	Noncontact
19	Presbyopic	Myope	Yes	Reduced	Noncontact
20	Presbyopic	Myope	Yes	Normal	Hard contact





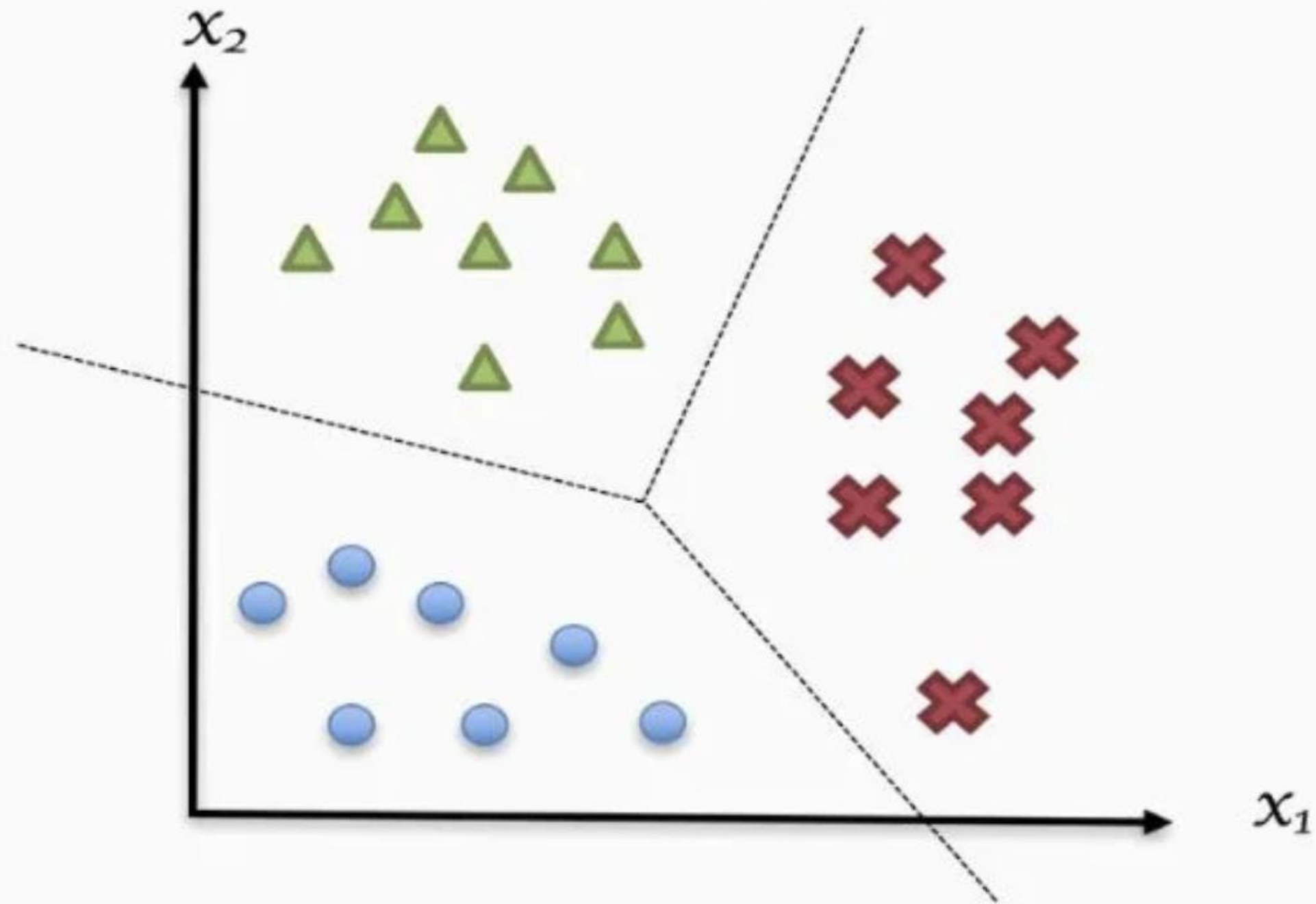
INBOX



SPAM FOLDER



Multiclass Classification



Multi-Class

Multi-Label

C = 3	Samples			Samples		
  						
	Labels			Labels		
	[100]	[010]	[001]	[110]	[011]	[111]

USE CASES



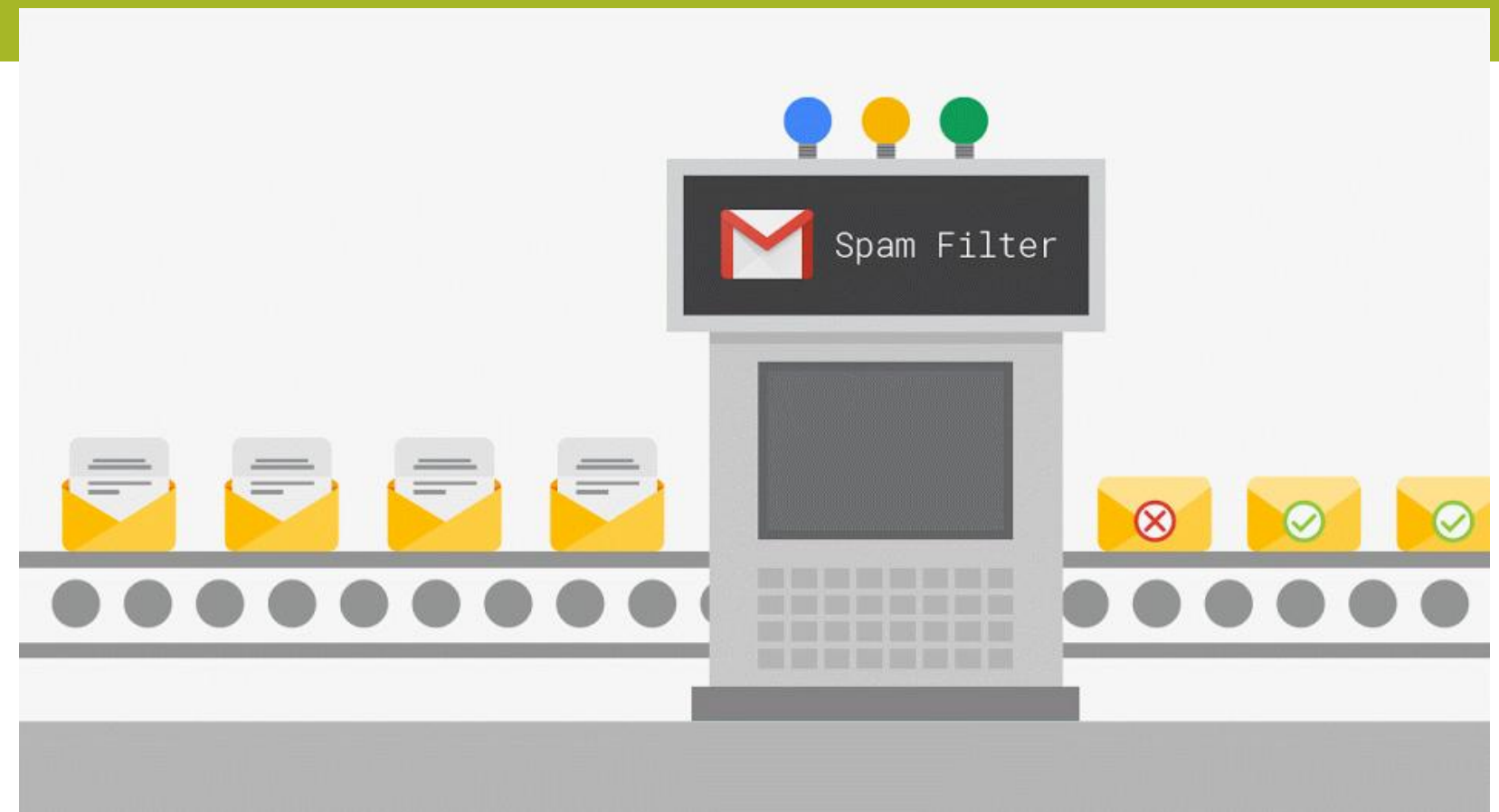
Andy Williams
CTO, Market Tech

Does anyone know when the shuttles leave this morning?

0 3



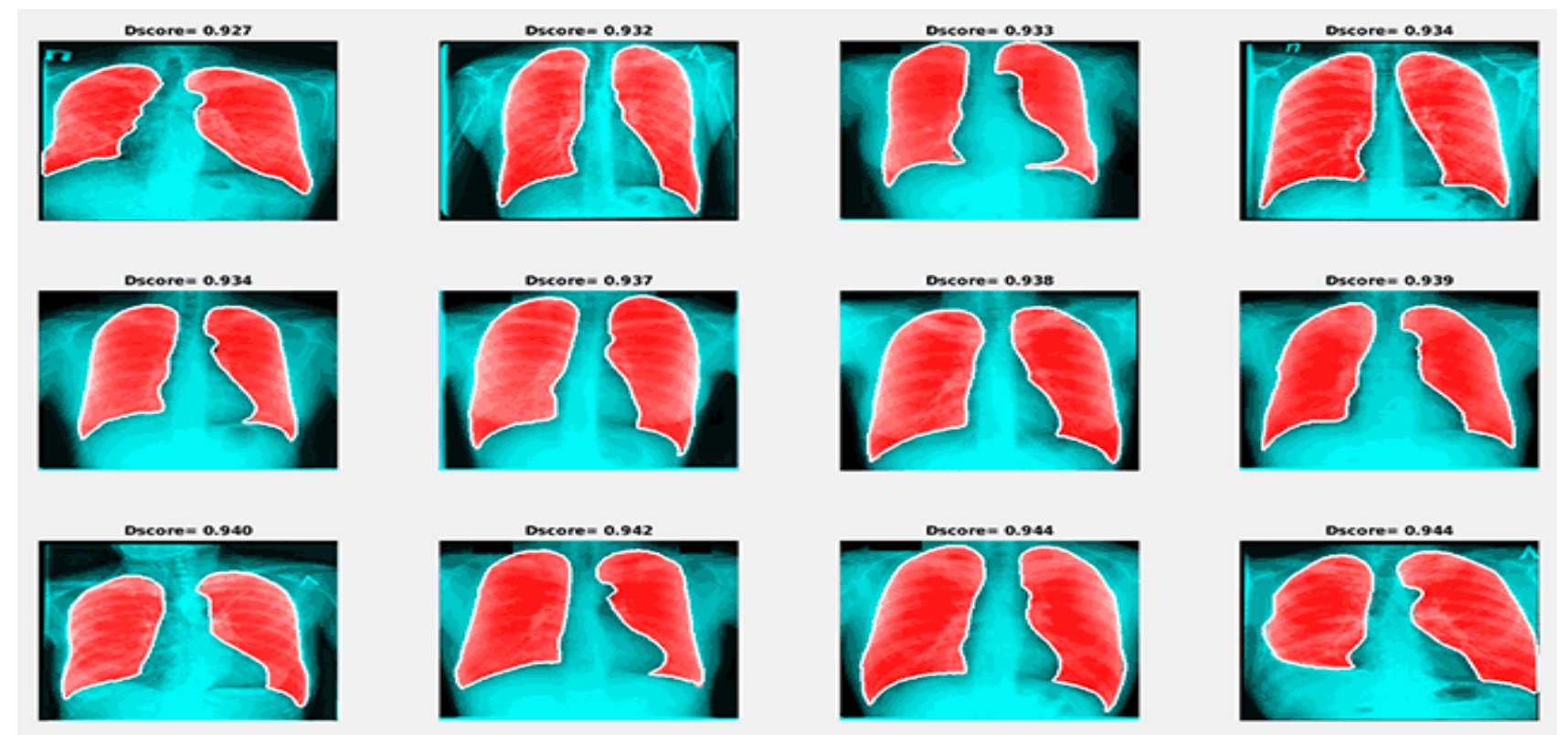
Sentiment Analysis



Spam Mail filtering

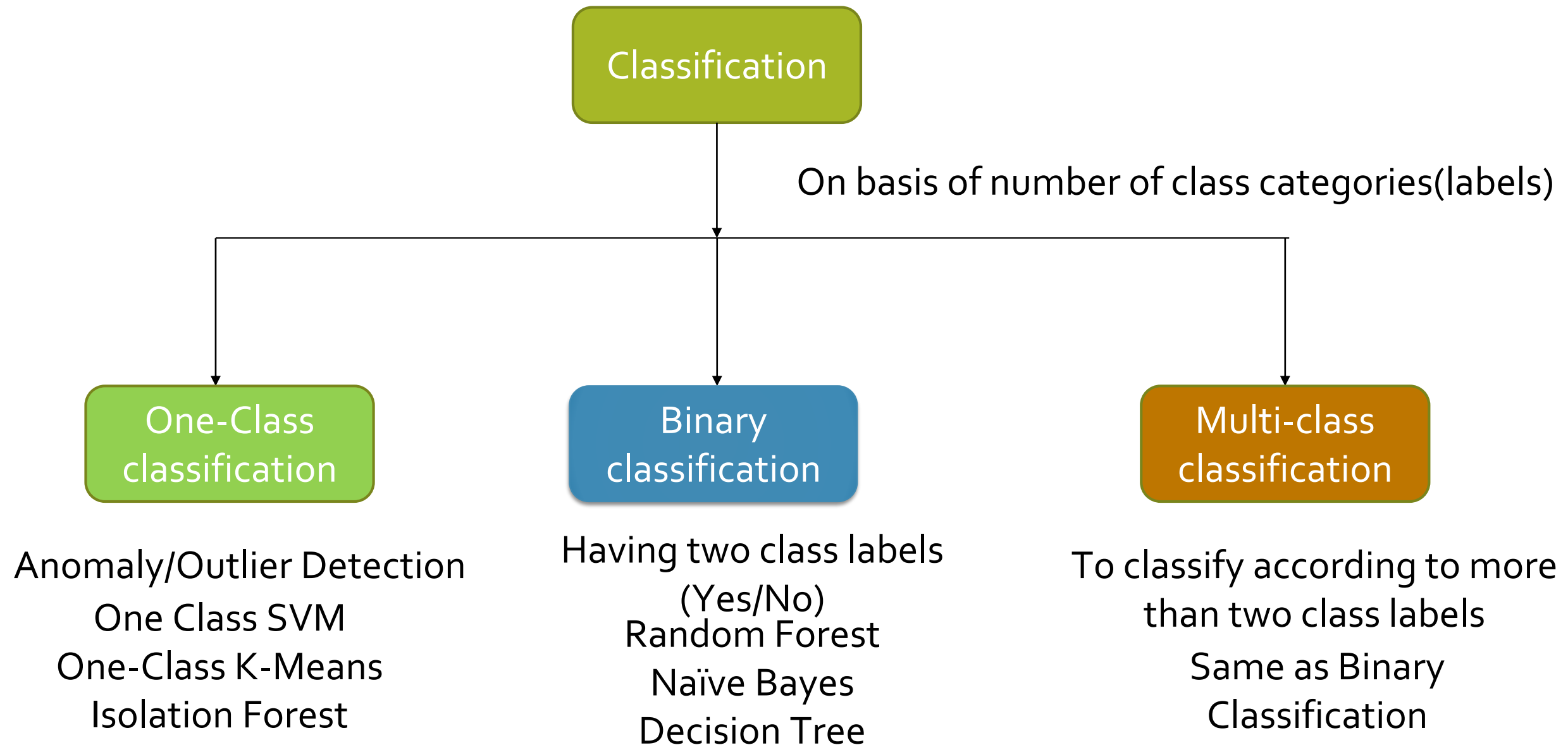


Plant disease classification



X-ray image segmentation

Classification types



What's the Difference Between a Classifier and a Model?

- a "classifier" refers specifically to the algorithm used to categorize data into different classes
- a "model" is the trained representation of that algorithm, which can then be used to make predictions on new data
- Pre-trained sentiment analysis model
- Email Intent Classifier
- Source: <https://monkeylearn.com/blog/what-is-a-classifier/>

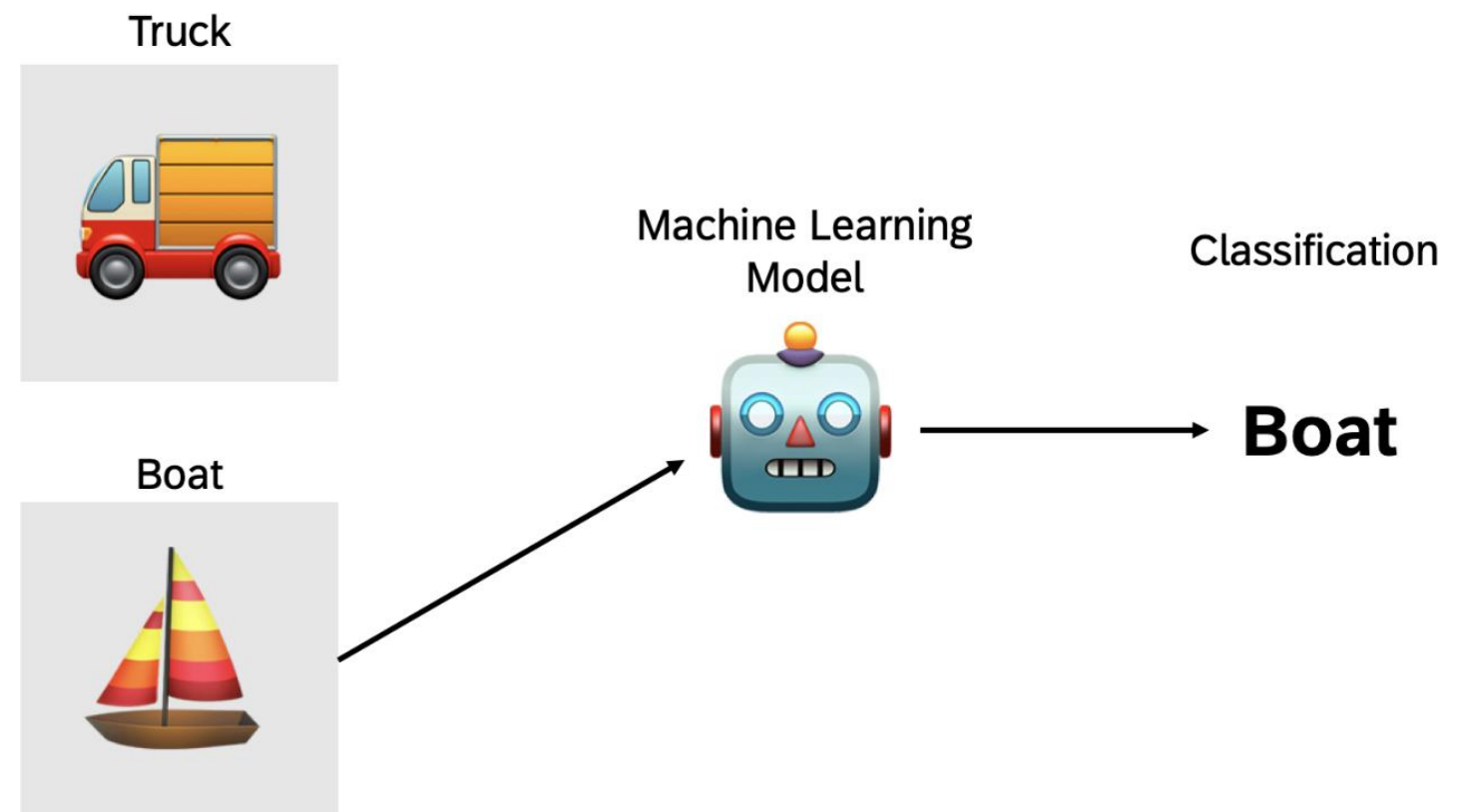
CLASSIFICATION TYPES IN MACHINE LEARNING

Python Demo

Link: <https://platoaistream.net/plato-data/a-complete-guide-to-understand-classification-in-machine-learning/>

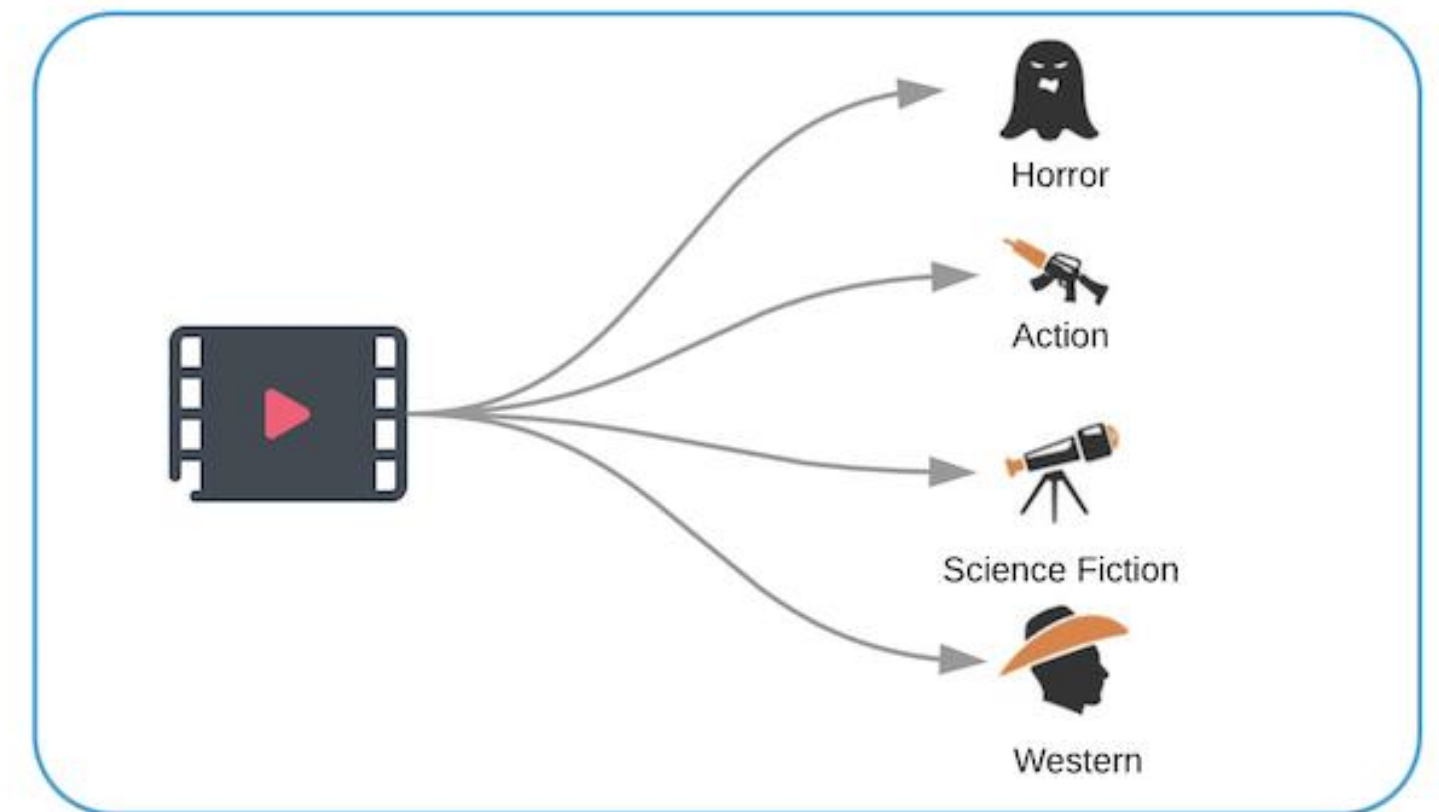
Binary Classification for Machine Learning

- A binary classification refers to those tasks which can give either of any two class labels as the output.
- The most popular algorithms which are used for binary classification are :
 - ❖ K-Nearest Neighbours
 - ❖ Logistic Regression
 - ❖ Support Vector Machine
 - ❖ Decision Trees
 - ❖ Naive Bayes



Multi-Class Classification

- These types of classification problems have no fixed two labels but can have any number of labels. Some popular examples of multi-class classification are :
 - Plant Species Classification
 - Face Classification
 - Optical Character recognition
- The most common algorithms which are used for Multi-Class Classification are :
 - ❖ K-Nearest Neighbours
 - ❖ Naive Bayes
 - ❖ Decision trees
 - ❖ Gradient Boosting
 - ❖ Random Forest



Multi-class classification

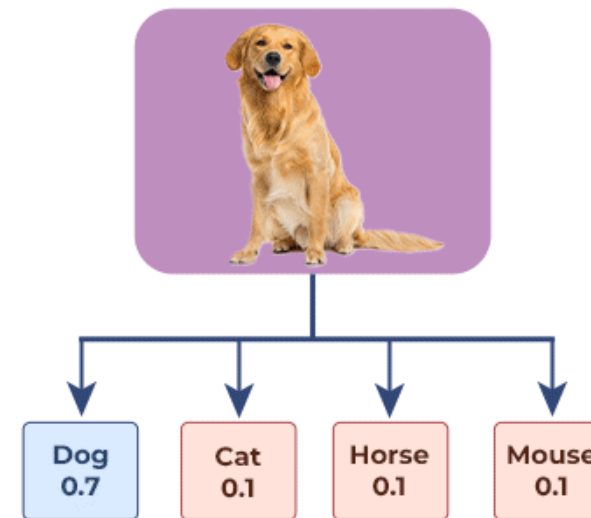
Multi-Label Classification for Machine Learning

- In multi-label Classification, we refer to those specific classification tasks where we need to assign two or more specific class labels that could be predicted for each example.
- ❖ Multi-label Random Forests
- ❖ Multi-label Decision trees
- ❖ Multi-label Gradient Boosting

Mutliclass Classification vs multilabel classification



Multiclass Classification

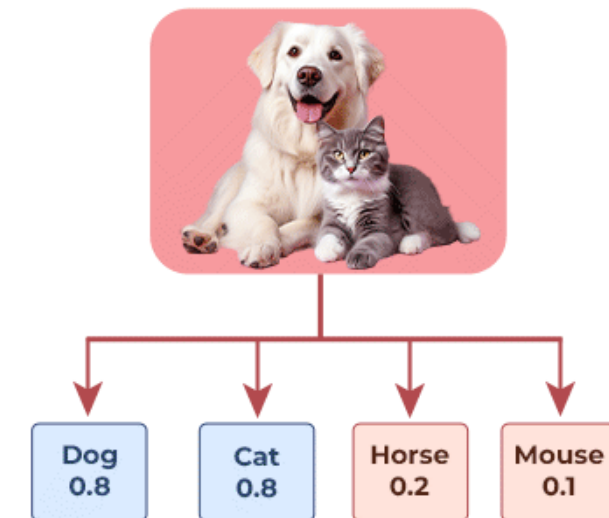


Classes

(pick one class)

- ☒ Dog
- ☐ Cat
- ☐ Horse
- ☐ Mouse

Multilabel Classification



Classes

(pick all the labels present in the image)

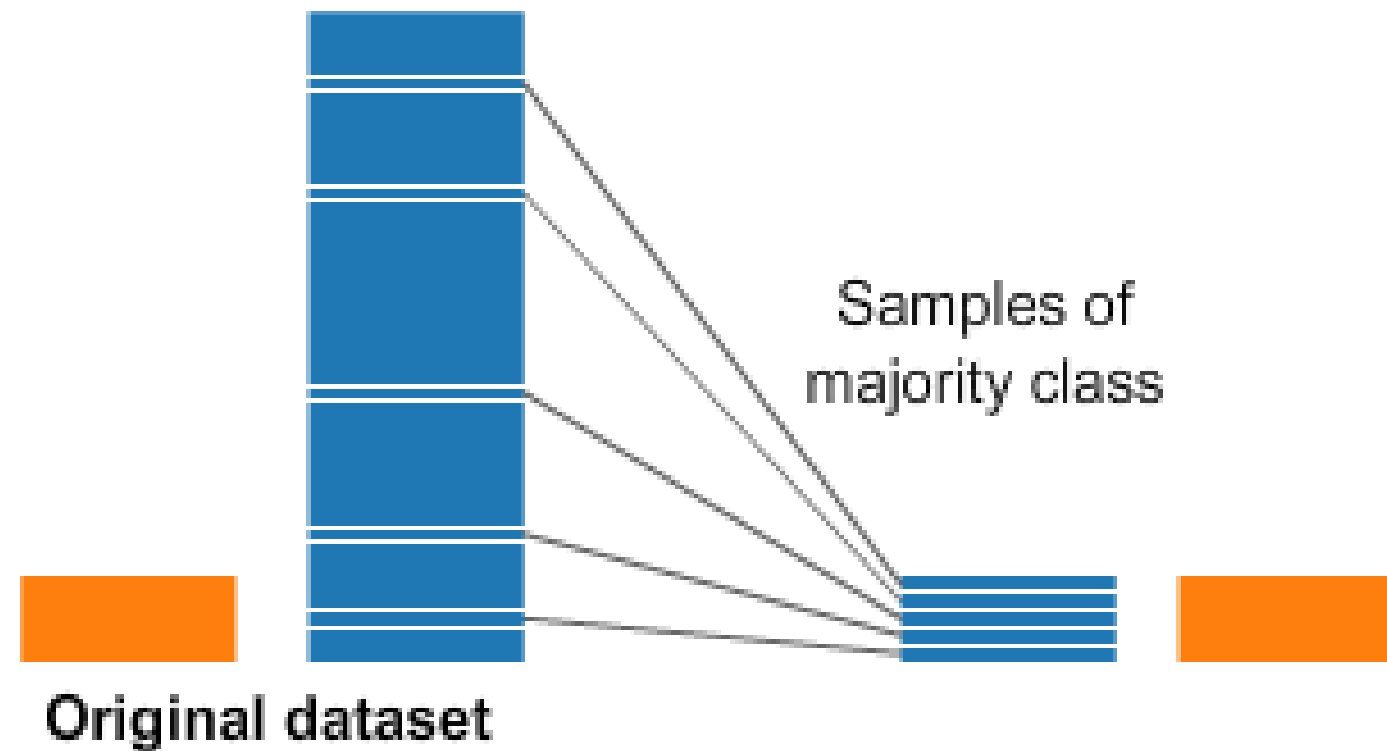
- ☒ Dog
- ☒ Cat
- ☐ Horse
- ☐ Mouse

Imbalanced Classification for Machine Learning

- An Imbalanced Classification refers to those tasks where the number of examples in each of the classes are unequally distributed. Generally, imbalanced classification tasks are binary classification jobs where a major portion of the training dataset is of the normal class type and a minority of them belong to the abnormal class.
- The most important examples of these use cases are :
 - Fraud Detection
 - Outlier Detection
 - Medical Diagnosis Test
- The problems are transformed into binary classification tasks with some specialized techniques. You can either utilise undersampling for the majority classes or oversampling for the minority classes. The most prominent examples are :
 - Random Undersampling
 - SMOTE Oversampling (Synthetic Minority Oversampling Technique)

Undersampling vs Oversampling

Undersampling



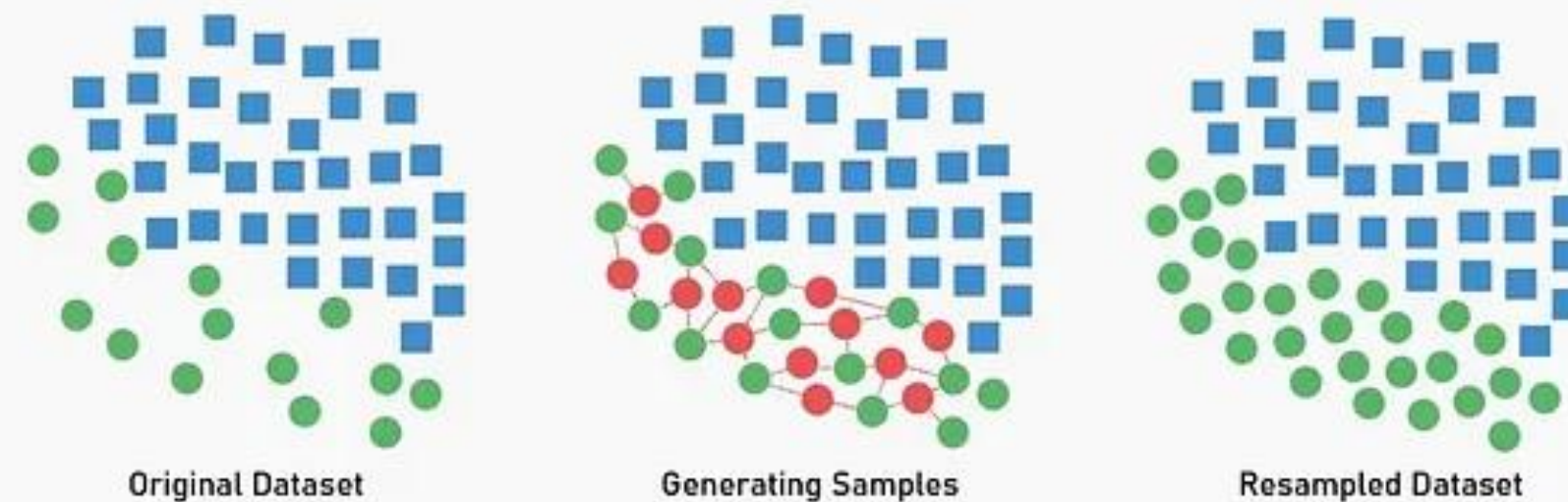
Oversampling



SMOTE

HANDLE IMBALANCED DATASET

Synthetic Minority Oversampling Technique



- 1. Selecting a Minority Class Instance:** Randomly choose a minority class instance, denoted as x .
- 2. Finding Nearest Neighbors:** Identify k nearest neighbors of x within the minority class.
- 3. Creating Synthetic Instances:** Randomly select one of the k nearest neighbors, denoted as x_{neighbor} , and generate a new synthetic instance x_{new} using the formula:

$$x_{\text{new}} = x + \delta \times (x_{\text{neighbor}} - x)$$

where δ is a random number between 0 and 1. By repeating this process, SMOTE generates a specified number of synthetic samples, increasing the size of the minority class.

Activity

Example	Prediction/Classification	Type
Filter Spam		
Purchasing Product X		
Producing revenue		
Failing in the manufacturing process		
Differing from known cases		
Defaulting on a loan		
Identify animal types in images from an encyclopedia		

Answers

Example	Prediction/Classification	Type
Filter Spam	Yes or No	Binary Classification
Purchasing Product X	Yes or No	Binary Classification
Producing revenue	Low, Medium, High	Multi-class Classification
Failing in the manufacturing process	Yes or No	Binary Classification
Differing from known cases	Yes or No	One-class Classification
Defaulting on a loan	Yes or No	Binary Classification
Identify animal types in images from an encyclopedia	Types of Animals	Multi-class Classification